

TrES-2b

R-1745

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_150831 · created 2026-08-02T19:08:32+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457265.78699 +/- 0.00039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.044
Transit depth (Rp/Rs)^2	2.22 +/- 1.31 [%]
Orbital Inclination (inc)	84.21 +/- 2.84
Airmass coefficient 1 (a1)	78.08 +/- 50.02
Airmass coefficient 2 (a2)	-2.73 +/- 0.52
Scatter in the residuals of the lightcurve fit is	45.82 %
Best Comparison Star	None
Optimal Aperture	3.61
Optimal Annulus	21.39
Transit Duration (day)	0.07 +/- 0.039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.044 vs 0.1254 (deviation 0.54σ)
Duration fit vs published	0.07 d vs 0.0746 d (deviation 0.12σ)
Mid-time O–C	0.2 min
Depth SNR	3.4
Residual scatter	45.82 %

Light curve

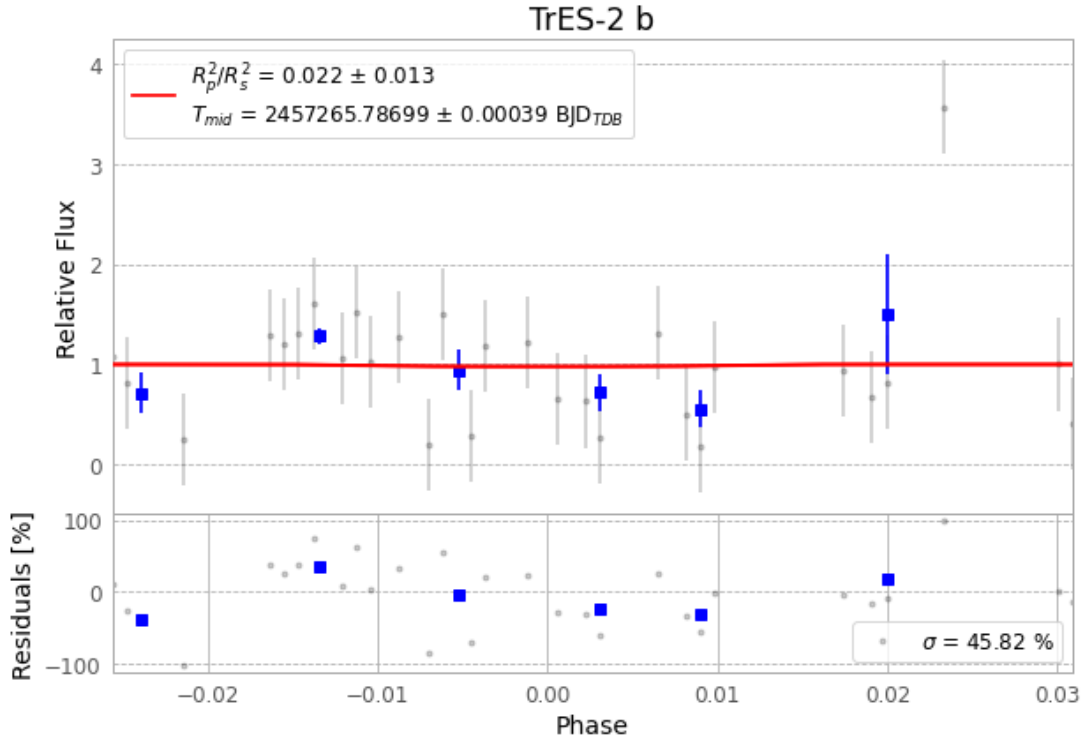


Figure 1 — FinalLightCurve_TrES-2 b_2015-08-30.png (SHA-256 457366313b0d08b4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [321, 432], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2d0c0a01dbbead59...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), TRES-2150831051212.FITS → TRES-2150831083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-150831.log (00a757162da0...), FinalLightCurve_TrES-2 b_2015-08-30.png (457366313b0d...), FinalLightCurve_TrES-2 b_2015-08-30.csv (80fab5136c02...)

Compute resources

Wall time 235.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 19:08Z.